

AI STUDYBUDDY API

Proposed Solution

Functional backend outline

TEAM ID	SWTID-2026-2776
TEAM SIZE	5
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Overview

Provide an authenticated API that stores users and study materials in MongoDB and calls Gemini to create reusable learning aids.

Modules

- index.js starts Express, middleware, route groups, health response, and database connection.
- controllers handle auth, material, and admin workflows.
- models define User and Material.
- middleware enforces JWT/role and handles file upload.
- utils provide database, Gemini, and token helpers.

Student operations

Authenticate, upload supported study file, manage own materials, and request four AI outputs.

Administrator operations

Review users, delete account and related materials, and check aggregate counts.

Design risk

Reconcile README cookie claims with actual JSON-token/Bearer implementation; enforce server-controlled roles and per-owner checks on every material operation.